# Solar Panel Installation Lead Management System

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# 1. Project Overview

## 1.1 Background

The company engaged in the installation and maintenance of solar panels across residential, industrial, agricultural, and other categories. The business relies on field executives who visit potential customer sites, assess feasibility, gather documentation, and facilitate the end-to-end installation process including financing assistance through loan processing.

## 1.2 Problem Statement

Currently, lead management is fragmented. There is no centralized system to capture leads from marketing channels, assign them intelligently to field executives, track progress across multiple stages, and communicate with the customer in a structured manner.

## 1.3 Objective

To build a unified platform consisting of:

- A public-facing lead capture landing page integrated with marketing channels.
- A web-based admin portal for centralized management of leads, users, and pipeline stages.
- A mobile application for field executives to manage assigned leads, collect customer data and documents, and process token payments on the field.

## 1.4 Scope

The scope of this project includes the following components:

- Lead capture form (publicly accessible)
- Web admin portal (role-based access)
- Field executive mobile application (iOS and Android)
- Automated lead assignment engine
- Multi-channel customer notification system (SMS, Email, WhatsApp)
- Document storage and management
- UPI and QR-based payment collection

- Real-time lead tracking and status management

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# 2. Stakeholders and Roles for Web Admin Portal

## 2.1 Role Definitions

**Super Admin**
Has complete access to the entire platform. Can create and manage all users including admins, district managers, and field executives. Can configure lead statuses, assignment rules, and system-wide settings. Has access to all reports and analytics.

**Admin**
Has access to the admin portal with the ability to manage leads, users within their assigned scope, and communications. Cannot modify system-level configurations unless explicitly granted by the Super Admin.

**District Manager**
Responsible for overseeing all leads and field executives within one or more assigned districts. Can view and manage leads assigned to their district. Can reassign leads between field executives under their management. Receives notifications for all lead activity within their district.

**Field Executive**
A ground-level employee who visits customers. Has access only to the mobile application. Can view their assigned leads, update lead status, upload documents, and collect token payments. Has no access to the web admin portal.

**Customer**
Not a system user. Submits the initial lead through the landing page and receives automated notifications via SMS, Email, and WhatsApp throughout the process.

## 2.2 Role Permissions Matrix

| Feature | Super Admin | Admin | District Manager | Field Executive |
|---|---|---|---|---|
| Create / Manage Users | Yes | Partial | No | No |
| Configure Lead Statuses | Yes | No | No | No |
| Configure Assignment Rules | Yes | No | No | No |
| View All Leads | Yes | Yes | District Only | Assigned Only |
| Reassign Leads | Yes | Yes | District Only | No |
| Approve / Reject Users | Yes | Yes | No | No |

| Feature | Super Admin | Admin | District Manager | Field Executive |
|---|---|---|---|---|
| View Reports | Yes | Yes | Partial | No |
| Upload Documents | Yes | Yes | Yes | Yes |
| Verify UTR | Yes | Yes | Yes | No |
| Manage Notifications | Yes | Yes | No | No |

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# 3. System Architecture Overview

## 3.1 High-Level Architecture

The system follows a three-tier architecture:

**Presentation Layer**
The landing page (Next.js), the web admin portal (Next.js + React), and the mobile application (React Native) form the presentation layer. These interact exclusively with the backend through RESTful APIs.

**Application Layer**
A Node.js backend exposes a structured REST API and handles business logic including lead assignment, status transitions, notification dispatching, and payment verification. Firebase is used for real-time data synchronization for notifications and live lead status updates.

**Data Layer**
A relational database (PostgreSQL) stores all structured data including users, leads, statuses, audit logs, and payment records. Amazon S3 is used for storing all uploaded documents.

## 3.2 Communication Flow

```
[Lead Source: Ad / Website]
         |
         v
[Landing Page - Lead Capture Form]
         |
         v
[Node.js Backend API]
         |
    +---------+
    |         |
    v         v
[PostgreSQL]  [Firebase]
```

```
     |         |
     |         v
     |   [Push Notifications to Admin Portal & Mobile App]
     |
     v
 [Auto-Assignment Engine]
          |
          v
 [Field Executive Mobile App]
          |
     +---------+-----------+
     |         |           |
     v         v           v
 [S3 Docs]  [Payment]  [Status Updates]
                           |
                           v
               [Customer Notifications: SMS, Email, WhatsApp]
```

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# 4. Lead Capture Landing Page

## 4.1 Page Requirements

The landing page must be a standalone Next.js page, optimized for SEO and Core Web Vitals. It must be independently deployable and must not require user authentication.

The page must include:

- A clear, compelling headline communicating the company's value proposition.
- Optionally, a brief description of the service including installation types (residential, industrial, agricultural).
- A prominently placed lead capture form.
- The company's contact information and official branding.
- Trust indicators such as certification logos, number of installations completed, and customer testimonials if available.

## 4.2 Lead Capture Form

**Form Fields:**

| Field | Type | Validation | Required |
|---|---|---|---|
| Full Name | Text | Min 2 characters, alphabets only | Yes |
| Phone Number | Tel | 10-digit Indian mobile number, numeric | Yes |

| Field | Type | Validation | Required |
|---|---|---|---|
| Email Address | Email | Valid email format | Yes |
| Monthly Electricity Bill (INR) | Number | Positive integer, min value configurable | Yes |
| District | Searchable Dropdown | Must match predefined district list | Yes |
| State | Read-only Text | Auto-populated based on district selection | Yes |
| Installation Type | Single Select | Residential, Industrial, Agricultural, Other | Yes |
| Message / Additional Info | Textarea | Max 500 characters | No |
| Consent Checkbox | Checkbox | Must be checked before submission | Yes |

**Form Behavior:**

- District field must support live search filtering. The district-to-state mapping must be maintained as a data table in the backend.
- State field is read-only and auto-populates instantly upon district selection without requiring an API call, using a locally bundled mapping.
- Real-time validation must occur on each field on blur and on form submission attempt.
- Duplicate lead detection: if a phone number already exists in the system with an active lead (not closed or rejected), a soft warning message should be displayed but submission must still be allowed to avoid blocking genuine re-inquiries.
- On successful submission, the form must be replaced with a success message thanking the user and informing them that a representative will contact them shortly.
- On failure, a generic error message must be displayed without revealing backend details.
- The form must be protected by Google reCAPTCHA v3 to prevent automated bot submissions.
- UTM parameters from the URL must be captured at submission and stored with the lead record for marketing attribution.

## 4.3 Performance Requirements

- The landing page must achieve a Lighthouse performance score of at least 90 on mobile.
- Time to Interactive (TTI) must not exceed 3 seconds on a 4G connection.
- The page must be statically generated (SSG) or server-side rendered (SSR) as appropriate to ensure SEO compliance.

# 5. Web Admin Portal

## 5.1 Authentication

**Login Mechanism:**
Email and password authentication only. No social login. No self-registration. Users must be created by a Super Admin or Admin and will receive a one-time setup email to set their password.

**Session Management:**
JWT-based authentication with access tokens and refresh tokens. Access tokens expire after 15 minutes. Refresh tokens expire after 7 days. Tokens are stored in HTTP-only cookies to prevent XSS attacks.

**Account Approval:**
All created user accounts default to a pending state. A Super Admin or Admin must explicitly approve the account before the user can log in. Suspended or deactivated accounts must be prevented from logging in even with valid credentials.

**Password Policy:**
Minimum 8 characters, must include at least one uppercase letter, one lowercase letter, one number, and one special character. Passwords must be hashed using bcrypt with a minimum work factor of 12.

**Multi-Factor Authentication:**
MFA via email OTP is recommended for Super Admin and Admin accounts and must be configurable.

**Audit Logging:**
All login attempts (successful and failed), password changes, and user management actions must be logged with timestamp, IP address, and actor identity.

## 5.2 Dashboard

The dashboard is the default landing page post-login. It provides a real-time summary of the system state.

**Dashboard Widgets:**

- Total leads captured today, this week, this month.
- Leads by current status (visual pipeline breakdown as a funnel or bar chart).
- Leads by district (geographic distribution).
- Leads by installation type.
- Field executive performance summary (leads assigned, visits completed, token amount collected).
- Pending verifications (UTR verification, document review).

- Recent activity feed showing the latest lead updates across the system.
- Loan application pipeline summary (applications pending, approved, rejected).

All widgets must be filterable by date range, district, and field executive. Data must refresh at a configurable interval (default 60 seconds) or on manual refresh.

## 5.3 Lead Management

**Lead List View:**

The lead list must be a paginated, searchable, and filterable data table. Default data is limited to 10 with pagination support. Default sort order is by creation date descending.

Filters available:

- Date range (lead creation date)
- Status (multi-select from configured status list)
- District (multi-select)
- State
- Assigned field executive
- Installation type
- Source (UTM source if available)

Search must work across lead name, phone number, and email.

Each row in the table must show: Lead ID, Name, Phone, District, Status, Assigned Executive, Creation Date, Last Updated Date, and quick action buttons.

**Lead Detail View:**

Clicking a lead opens a detailed view structured in tabs or sections:

- **Overview:** Basic lead information as captured from the form. Includes UTM source/medium/campaign data.
- **Status Timeline:** A chronological visual timeline of all status transitions including who changed the status, when, and any notes attached to the transition.
- **Customer Information:** Detailed customer data filled by the field executive during the site visit.
- **Documents:** All uploaded documents with preview capability and download option. Documents must be categorized by type.
- **Payment:** Token payment status, UTR number if applicable, and verification status.
- **Loan Details:** Loan application status, loan amount, lender name, disbursement date, and other relevant fields.
- **Communications:** Log of all notifications sent to the customer (SMS, Email, WhatsApp) with delivery status where available.

- **Notes:** Internal notes added by admin users or managers. Not visible to customers or field executives.
- **Activity Log:** Complete audit trail of all actions taken on this lead.

**Lead Assignment:**

Leads can be assigned or reassigned from the lead detail view. The assignment interface must show available field executives for the relevant district with their current active lead count to assist in workload balancing.

Manual reassignment must require a reason to be entered and must trigger notifications to the newly assigned executive and the previously assigned executive.

## 5.4 Lead Status Configuration

This is a configurable workflow engine. Super Admins can define the complete lead status pipeline.

**Status Management Interface:**

- A visual, ordered list of all statuses representing the lead lifecycle.
- Ability to add new statuses, edit existing ones, reorder them, and mark them as terminal (closed, rejected, installation complete).
- For each status, the following must be configurable:
  - Status name and description
  - The list of statuses that can follow it (allowed next transitions)
  - Whether this transition requires a note
  - Whether this transition triggers a customer notification and which notification template to use
  - Whether this transition requires document upload
  - Color coding for visual differentiation in the UI

**Default Status Sequence:**

The system must ship with the following default statuses pre-configured, which can be modified by the Super Admin:

1. New (initial state on lead capture)
2. Assigned
3. Field Visit Scheduled
4. Field Visit Done
5. Documents Uploaded by Field Executive
6. Token Amount Received (INR 1000)
7. Token Payment Verification Pending
8. Token Payment Verified

9. Verification Pending (internal verification of customer eligibility)
10. Verified
11. Loan Application Initiated
12. Loan Approval Pending
13. Loan Disbursed
14. Loan Rejected
15. Installation Scheduled
16. Installation In Progress
17. Installation Complete
18. Closed (Lost)

Statuses 17 and 18 are terminal statuses. Once a lead reaches a terminal status, it cannot be moved to another status unless a Super Admin overrides it.

## 5.5 User Management

**User List:**
A paginated, searchable, and filterable table of all system users. Filters include role, status (active, pending, suspended), and district.

**Create User:**
A form to create a new user with fields: Full Name, Email, Role, Assigned District(s), Phone Number, Employee ID (optional). On creation, the system sends a setup email with a secure link to set a password.

**Edit User:**
Modify user details, role, or district assignment. Role downgrades must log a warning and require confirmation.

**Approve / Suspend / Deactivate User:**
Explicit actions with confirmation dialogs. Deactivating a user with active lead assignments must prompt the admin to reassign those leads.

**Field Executive Profile View:**
Shows complete profile of a field executive including all leads assigned, current active leads, completion rate, token collections, and document submission record.

## 5.7 District and Geography Management

**District Management:**
A configurable master list of districts and their associated states. The Super Admin can add, edit, or deactivate districts. Each district must have at least one assigned District Manager before leads from that district can be auto-assigned.

**District-Manager Mapping:**
Interface to assign one or more District Managers to each district.

**District-Executive Mapping:**
Interface to assign one or more Field Executives to each district. A field executive can be assigned to multiple districts.

## 5.8 Document Review

A dedicated section to review documents uploaded by field executives. Admin users can:

- Preview documents in-browser (PDF, JPEG, PNG supported).
- Download individual documents.
- Mark a document as verified or request re-upload with a reason.
- Add notes against individual documents.

Document categories to be reviewed:

- Electricity Bill
- Aadhaar Card (Front and Back)
- PAN Card
- Bank Account Details / Cancelled Cheque
- Site Photographs
- Roof Assessment Report
- Any additional documents as configured

## 5.9 Payment Verification

A dedicated queue for UTR verification. Each entry shows:

- Lead ID and customer name
- Amount claimed
- UTR number entered by the field executive
- Date and time of submission
- Field executive name

Admin or District Manager can mark as verified or rejected with a note. On verification, the lead status automatically transitions to Token Payment Verified. On rejection, the field executive receives a notification to retry.

## 5.10 Notification Management

**Notification Templates:**
Admin can manage SMS, Email, and WhatsApp notification templates. Templates support dynamic variables such as customer name, lead ID, field executive name, and current status. Templates are associated with specific lead status transitions.

**Notification Logs:**
A complete log of all customer notifications sent with delivery status, channel, template used,

timestamp, and associated lead.

**Internal Notifications:**
Admin portal must show real-time browser push notifications for: new lead submissions, UTR submissions pending verification, document uploads pending review, and any overdue leads (leads that have not changed status beyond a configurable threshold period).

## 5.11 Reports and Analytics

The reporting section must provide the following reports, all exportable to CSV and PDF:

- Lead Source Report: Leads by UTM source, medium, and campaign over a selected date range.
- Lead Pipeline Report: Conversion rates between each stage of the pipeline.
- District Performance Report: Leads by district with conversion metrics.
- Field Executive Performance Report: Individual executive metrics including leads assigned, visits completed, documents submitted, and token collections.
- Revenue Report: Token amounts collected by period and by district.
- Loan Pipeline Report: Loan applications by status, lender, and district.
- Customer Communication Report: Notification delivery rates by channel.

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# 6. Mobile Application for Field Executives

## 6.1 Authentication

- Email and password login, same credential system as the web portal.
- Biometric authentication (fingerprint or face unlock) must be supported as a secondary authentication method after the initial login.
- Sessions persist until explicitly logged out or the refresh token expires.
- If an account is suspended or deactivated, the session must be terminated immediately and the user must see an informative message on next app open.

## 6.2 Home Screen / Dashboard

The home screen displays:

- A summary of the executive's active leads grouped by urgency or status.
- Today's tasks or visits scheduled.
- Count of pending actions (documents to upload, payments to collect, forms to complete).
- Recent notifications from the system.

## 6.3 Lead Management

**Lead List:**
A scrollable list of all leads assigned to the field executive. Each item shows the customer name, district, current status, and the last update date. The list must support filtering by status and sorting by date.

**Lead Detail:**
Tapping a lead opens the full detail view including:

- Customer contact information with a tap-to-call button.
- Google Maps integration with the customer address to open navigation directly.
- Current status with the allowed next statuses.
- Status update button to progress the lead to the next stage.
- Document upload section.
- Payment collection section.
- Notes field for internal notes visible only to the executive and admin.

**Status Update:**
When updating the status, the application must:

- Show only the next allowed statuses as per the configured workflow.
- Prompt for required notes if the transition mandates it.
- Prompt for document upload if the transition mandates it.
- Confirm the action before committing.

## 6.4 Customer Data Collection Form

When the field executive conducts a site visit, they must fill in a comprehensive customer data form. This form is associated with the lead and is submitted once and can be edited until the lead reaches a terminal status.

**Customer Information Section:**

| Field | Type | Required |
|---|---|---|
| Full Name | Text | Yes |
| Date of Birth | Date Picker | Yes |
| Gender | Single Select | Yes |
| Father / Husband Name | Text | Yes |
| Aadhaar Number | Numeric, 12 digits | Yes |
| PAN Number | Alphanumeric, 10 chars | Yes |
| Complete Address (House No., Street) | Text | Yes |
| Village / Locality | Text | Yes |
| Pin Code | Numeric, 6 digits | Yes |

| Field | Type | Required |
|---|---|---|
| District | Pre-filled from lead | Yes |
| State | Pre-filled from lead | Yes |
| Alternate Phone Number | Tel | No |

**Property and Installation Details Section:**

| Field | Type | Required |
|---|---|---|
| Installation Type | Single Select (Residential, Industrial, Agricultural, Other) | Yes |
| Property Ownership | Single Select (Owned, Rented, Leased) | Yes |
| Total Roof / Land Area Available (sq ft) | Numeric | Yes |
| Recommended Solar Capacity (kW) | Numeric | Yes |
| Shadow-Free Area Available | Single Select (Yes, Partial, No) | Yes |
| Type of Roof | Single Select (RCC, Tin, Other) | Yes |
| Current Monthly Electricity Bill (Verified) | Numeric | Yes |
| Current Electricity Connection Type | Single Select (Single Phase, Three Phase) | Yes |
| Electricity Consumer Number | Text | Yes |
| DISCOM / Electricity Board Name | Text | Yes |
| Site Photographs | Camera / Gallery Upload (min 3, max 10) | Yes |

**Financial and Bank Details Section:**

| Field | Type | Required |
|---|---|---|
| Bank Account Number | Numeric | Yes |
| Bank Name | Text | Yes |
| IFSC Code | Alphanumeric | Yes |
| Account Holder Name | Text | Yes |
| Loan Required | Boolean | Yes |
| Loan Amount Required (INR) | Numeric | If loan required |
| Preferred Lender (if any) | Text | No |

**Form Behavior:**

- The form must save progress locally at every field change to prevent data loss.
- The form must validate each field before allowing submission.
- Aadhaar number must be masked in display after entry (showing only last 4 digits).
- PAN number must be converted to uppercase automatically.
- IFSC code lookup must trigger the bank name and branch to auto-populate using a bank IFSC API.

## 6.5 Document Upload

The field executive must be able to upload documents directly from the mobile device using the device camera or the file system.

**Document Categories:**

- Aadhaar Card Front
- Aadhaar Card Back
- PAN Card
- Electricity Bill (latest)
- Cancelled Cheque or Bank Passbook Front Page
- Site Photographs (multiple)
- Roof Assessment or Site Plan (optional)

**Upload Behavior:**

- Each document upload goes directly to Amazon S3 via a pre-signed URL generated by the backend. The mobile application must not store AWS credentials.
- Supported file formats: JPEG, PNG, PDF.
- Maximum file size per document: 10 MB.
- The upload progress must be displayed to the user.
- If a document upload fails, the application must allow retry without re-selecting the file.
- Uploaded documents must be immediately visible in the web admin portal.
- The application must support offline queuing: if a document is selected for upload while offline, it must upload automatically when connectivity is restored.

## 6.6 Offline Functionality

The mobile application must support partial offline functionality:

- All assigned leads and their current data must be cached locally.
- The field executive must be able to view leads, customer details, and documents already downloaded while offline.
- The customer data collection form must be fillable offline with local save.

- Status updates and form submissions made while offline must be queued and synced when connectivity is restored.
- Document uploads must queue offline and upload when connectivity is restored.
- The application must display a clear offline indicator to the user.

### 6.7 Notifications

The mobile application must receive push notifications via Firebase Cloud Messaging (FCM) for the following events:

- New lead assigned to the field executive.
- Lead status updated by an admin or manager affecting the executive.
- UTR rejected by admin (requesting retry from the executive).
- Document rejected by admin (requesting re-upload).
- Reminder notification for leads that have not had any activity for a configurable number of days.
- Loan status update for any of the executive's leads.

Notification must deep-link to the relevant lead detail screen when tapped.

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## 7. Lead Lifecycle and Status Management

### 7.1 Status Transition Rules

Each lead always has exactly one current status. A status transition is valid only if the target status appears in the allowed transitions list configured for the current status. Any attempt to move to a non-allowed status must be rejected with an informative error.

Every status transition must be recorded in the lead's audit log with:

- Previous status
- New status
- Timestamp
- Actor (user ID and name)
- Notes (if provided or required)

### 7.2 Assignment-Triggered Transitions

When a lead is newly created, its status is automatically set to New. Upon successful auto-assignment to a field executive, the status must automatically transition to Assigned. If auto-assignment fails (no executive available in the district), the status remains New and an alert is raised to the District Manager and Admin.

### 7.3 Terminal Statuses

Terminal statuses are statuses from which no further transitions are allowed by default. Installation Complete and Closed (Lost) are terminal. Only a Super Admin can reopen a lead from a terminal status, which must log an override reason.

### 7.4 SLA and Overdue Detection

Each status must have a configurable SLA duration (for example, the lead must move out of Assigned status within 48 hours). If a lead stays in any non-terminal status beyond its configured SLA, the system must:

- Flag the lead as overdue in the admin portal.
- Send an internal notification to the District Manager and Admin.
- Display the lead with a visual indicator in the lead list.

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## 8. Auto-Assignment Engine

### 8.1 Assignment Logic

When a new lead is submitted, the auto-assignment engine executes the following logic:

1. Extract the district from the lead form submission.
2. Identify the District Manager assigned to that district.
3. Identify all active Field Executives assigned to that district.
4. Among available Field Executives, select the one with the lowest number of currently active (non-terminal) leads.
5. Assign the lead to the selected executive and update the status to Assigned.
6. Send notifications to the assigned executive, the District Manager, and Admin.

### 8.2 Assignment Failure Handling

If no Field Executive is available for the district (no executives assigned or all are at maximum load):

- The lead is assigned to the District Manager directly with a flag indicating no executive is available.
- An alert notification is sent to all Admin users.
- The lead remains in the New status until manual assignment.

### 8.3 Configurable Maximum Load

Super Admin must be able to configure the maximum number of active leads allowed per field executive. Once an executive reaches this limit, they are excluded from auto-assignment until their active lead count drops below the threshold.

## 8.4 Manual Override

Super Admin or Admin or District Manager can manually assign or reassign any lead at any point. Manual assignment overrides the auto-assignment result. All reassignments must be logged.

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# 9. Notification System

## 9.1 Internal Notifications (Web Portal and Mobile App)

Internal notifications are delivered via Firebase Cloud Messaging and displayed as browser push notifications on the web portal and push notifications on the mobile application.

**Web Portal Notification Triggers:**

- New lead submitted.
- New UTR payment submitted by a field executive.
- New documents uploaded by a field executive.
- Lead status updated.
- Field executive has not updated a lead within SLA period.

**Mobile App Notification Triggers:**

Documented in Section 6.8 above.

## 9.2 Customer Notifications

Customer notifications are multi-channel: SMS, Email, and WhatsApp. Each notification is triggered by a lead status transition and uses the template configured for that transition in the admin portal.

**Channels:**

- SMS: Delivered via a third-party SMS gateway (MSG91). Messages must be DLT-registered in compliance with TRAI regulations for India.
- Email: Delivered via a transactional email provider (recommended: SendGrid or AWS SES or Resend). Emails must include company branding and an unsubscribe option.
- WhatsApp: Delivered via WhatsApp Business API (recommended: Twilio or a BSP like Interakt or WATI). Messages must use pre-approved WhatsApp Business message templates.

**Notification Triggers and Default Templates:**

| Event | Channels |
|---|---|
| Lead form submitted | SMS, Email, WhatsApp |
| Lead assigned to field executive | SMS, WhatsApp |
| Field visit scheduled (if applicable) | SMS, WhatsApp |
| Field visit completed | SMS |
| Documents submitted by executive | SMS |
| Token payment received | SMS, Email, WhatsApp |
| Token payment verified | SMS, Email, WhatsApp |
| Loan application initiated | SMS, Email, WhatsApp |
| Loan approved (disbursed) | SMS, Email, WhatsApp |
| Loan rejected | SMS, Email, WhatsApp |
| Installation scheduled | SMS, WhatsApp |
| Installation complete | SMS, Email, WhatsApp |

**Delivery Tracking:**

The system must track delivery status for each notification where the provider API supports it (delivered, failed, read if available). All notification logs must be accessible in the admin portal.

**Retry Logic:**

Failed notifications must be retried up to 3 times with exponential backoff before being marked as permanently failed and flagged for manual review.

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# 10. Payment Integration

## 10.1 Token Payment Overview

A token amount of INR 1,000 is collected from the customer at the time of confirming intent to proceed. This payment is collected by the field executive on-site. Two methods are supported: UPI payment and QR code display.

## 10.2 UPI Payment Method

The mobile application integrates with a UPI payment gateway (recommended: Razorpay or PayU) to initiate a UPI collect request. The flow is:

1. Field executive enters the customer's UPI ID or selects UPI as payment method.
2. The application calls the backend to create a payment order.

3. The customer receives a UPI collect request on their UPI app (BHIM, GPay, PhonePe, etc.).
4. On payment success, the gateway webhook notifies the backend.
5. The backend updates the lead status to Token Amount Received and logs the payment details.
6. Customer and field executive receive confirmation notifications.

## 10.3 QR Code Payment Method

If the customer does not have a UPI app linked or prefers scanning a QR code:

1. The field executive selects QR code payment in the application.
2. The application displays the company's official UPI QR code.
3. The customer scans and pays INR 1,000.
4. The field executive enters the UTR (Unique Transaction Reference) number from the customer's payment confirmation.
5. The lead is moved to Token Payment Verification Pending status.
6. An admin or district manager verifies the UTR in the web portal.
7. On verification, the lead moves to Token Payment Verified. On rejection, the executive is notified to retry.

## 10.4 Payment Records

All payment records must store:

- Lead ID
- Payment method (UPI gateway or QR-UTR)
- Amount collected
- Transaction ID (from gateway) or UTR number
- Timestamp
- Collection status (pending, verified, rejected)
- Verified by (user ID, if UTR method)
- Verification timestamp

## 10.5 Payment Security

Payment gateway integration must use the server-side API and not expose secret keys to the client application. All payment callbacks must be verified using webhook signatures. No card or sensitive payment data is stored in the application database.

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# 11. Document Management

## 11.1 Storage

All documents are stored on Amazon S3 in a private bucket. Direct public access to the bucket must be disabled. All access to documents must go through pre-signed URLs generated by the backend that expire after a configurable duration (default 15 minutes).

## 11.2 Folder Structure

Documents are organized in S3 using the following key structure:

```
leads/{lead_id}/documents/{category}/{timestamp}_{filename}
```

## 11.3 Upload Flow

1. The client (mobile app or web portal) requests a pre-signed upload URL from the backend, specifying file type and size.
2. The backend validates the request and generates a pre-signed S3 PUT URL.
3. The client uploads directly to S3 using the pre-signed URL.
4. On upload completion, the client notifies the backend with the S3 key.
5. The backend records the document metadata in the database.

## 11.4 Document Versioning

If a document of the same category is re-uploaded (for example, following a rejection), the previous version must be retained in S3 with a versioned key. The system must present the latest version by default while allowing admin to view previous versions.

## 11.5 File Type and Size Validation

File type must be validated both on the client side (extension check) and on the server side (MIME type check). Maximum file size must be enforced at the pre-signed URL generation step using S3 content-length restrictions.

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# 12. Customer Communication

## 12.1 Consent Management

The lead capture form must include a clear consent statement confirming that the customer agrees to receive communications via SMS, Email, and WhatsApp regarding their solar installation inquiry. Consent must be recorded with the lead record including the timestamp and IP address.

## 12.2 Opt-Out Handling

Customers must be able to opt out of WhatsApp and Email communications by responding with STOP or clicking an unsubscribe link. Opt-out status must be recorded in the system and respected for all future communications. SMS DND status must be checked before sending promotional messages in compliance with TRAI regulations.

---

# 13. Reporting and Analytics

## 13.1 Dashboard Analytics

Real-time dashboard analytics must be provided as described in Section 5.3. Underlying data aggregation must use database-level queries with appropriate indexing to ensure dashboards load within 3 seconds.

## 13.2 Export Functionality

All reports must be exportable in CSV format and PDF format. CSV exports must be generated on the server to handle large datasets. PDF exports must be formatted with company branding, report title, date range, and page numbers.

---

# 14. Security Requirements

## 14.1 Authentication and Authorization

All API endpoints must require authentication except the lead submission endpoint and the district-state mapping endpoint. Authorization checks must be enforced at the route handler level and must not rely solely on frontend routing.

## 14.2 API Security

- All API communication must occur over HTTPS (TLS 1.2 or higher).
- API rate limiting must be enforced, especially on the lead submission endpoint (recommended: 5 requests per IP per minute to prevent spam).
- CORS must be configured to allow only known client origins.
- All input must be validated and sanitized on the server side before processing or storage.
- SQL injection and NoSQL injection must be prevented through the use of parameterized queries and ORM-level validation.
- OWASP Top 10 vulnerabilities must be addressed in the implementation.

## 14.3 Data Privacy

- Sensitive customer data such as Aadhaar number, PAN number, and bank account number must be encrypted at rest in the database using AES-256.

- Sensitive fields must be masked in API responses unless the requesting user has the appropriate role.
- All data transfers between services must occur over encrypted channels.

### 14.4 Secret Management

All secrets including database credentials, S3 access keys, payment gateway keys, and Firebase service account credentials must be stored in environment variables and managed using a secret management service (recommended: AWS Secrets Manager or HashiCorp Vault). No secrets must be committed to version control.

---

## 15. Technology Stack

### 15.1 Web Admin Portal and Landing Page

| Component | Technology |
|---|---|
| Frontend Framework | Next.js (React) |
| UI Library | React with Tailwind CSS |
| State Management | React Context API or Zustand |
| Form Management | React Hook Form with Zod validation |
| Data Fetching | Axios or SWR |
| Authentication State | JWT in HTTP-only cookies |
| Real-time Notifications | Firebase Cloud Messaging / Firestore |
| Charts and Analytics | Recharts or Chart.js |

### 15.2 Backend API

| Component | Technology |
|---|---|
| Runtime | Node.js (v20 LTS or higher) |
| Framework | Express.js or Fastify |
| ORM | Prisma |
| Database | PostgreSQL |
| File Storage | Amazon S3 |
| Authentication | JWT (jsonwebtoken library) |
| Email Service | SendGrid or AWS SES |
| SMS Service | MSG91, Twilio, or Exotel |
| WhatsApp API | Twilio or Interakt / WATI |

| Component | Technology |
|---|---|
| Payment Gateway | Razorpay or PayU |
| Push Notifications | Firebase Cloud Messaging (FCM) |
| Task Queue | Bull (Redis-backed) for async jobs |
| Caching | Redis |

## 15.3 Mobile Application

| Component | Technology |
|---|---|
| Framework | React Native (Expo managed workflow or bare workflow) |
| Navigation | React Navigation |
| State Management | Zustand or Redux Toolkit |
| Form Management | React Hook Form |
| Document Upload | expo-document-picker, expo-image-picker |
| Push Notifications | Firebase Cloud Messaging via @react-native-firebase |
| Offline Storage | WatermelonDB or AsyncStorage |
| Maps Integration | React Native Maps (Google Maps) |
| Biometric Auth | expo-local-authentication |

## 15.4 Infrastructure and DevOps

| Component | Technology |
|---|---|
| Version Control | Git with GitHub |
| CI/CD | GitHub Actions |
| Hosting (Frontend) | Vercel or AWS Amplify |
| Hosting (Backend) | AWS EC2 or AWS ECS (Docker containers) |
| Database Hosting | AWS RDS (PostgreSQL) |
| File Storage | Amazon S3 with CloudFront CDN |
| Caching Layer | AWS ElastiCache (Redis) |
| Monitoring | AWS CloudWatch or Datadog |
| Error Tracking | Sentry |
| Secret Management | AWS Secrets Manager or dotenv with restricted access |

# 16. Database Design (Summary, Design as per Actuals)

## 16.1 Core Tables

### users

```
id, email, password_hash, full_name, phone, role, employee_id, status
(active, pending, suspended),
created_at, updated_at, last_login_at
```

### districts

```
id, name, state, is_active, created_at
```

### user_district_assignments

```
id, user_id, district_id, assigned_at
```

### lead_statuses

```
id, name, description, order_index, is_terminal, color_code, requires_note,
requires_document,
notify_customer, notification_template_id, created_at, updated_at
```

### lead_status_transitions

```
id, from_status_id, to_status_id (defines the allowed workflow graph)
```

### leads

```
id, external_id (UUID for customer reference), name, phone, email,
monthly_bill,
district_id, state, installation_type, message, current_status_id,
assigned_executive_id,
assigned_manager_id, utm_source, utm_medium, utm_campaign, utm_term,
utm_content,
source_ip, recaptcha_score, consent_given, consent_timestamp, is_overdue,
created_at, updated_at
```

### lead_status_history

```
id, lead_id, from_status_id, to_status_id, changed_by_user_id, notes,
created_at
```

### customer_details

```
id, lead_id, full_name, date_of_birth, gender, father_husband_name,
aadhaar_encrypted,
pan_encrypted, address_line1, address_line2, village_locality, pincode,
district_id,
alternate_phone, property_ownership, roof_area, recommended_capacity,
shadow_free_area,
roof_type, verified_monthly_bill, connection_type, consumer_number,
discom_name,
bank_account_encrypted, bank_name, ifsc_code, account_holder_name,
loan_required,
loan_amount_required, preferred_lender, created_at, updated_at
```

### documents

```
id, lead_id, category, s3_key, file_name, file_type, file_size, version,
is_latest,
uploaded_by_user_id, review_status (pending, verified, rejected),
review_notes,
reviewed_by_user_id, reviewed_at, created_at
```

### payments

```
id, lead_id, amount, method (upi_gateway, qr_utr), gateway_order_id,
gateway_payment_id,
utr_number, status (pending, verified, rejected), collected_by_user_id,
verified_by_user_id, rejection_reason, created_at, verified_at
```

### notification_templates

```
id, name, channel (sms, email, whatsapp), subject, body_template, is_active,
created_at, updated_at
```

### notification_logs

```
id, lead_id, channel, template_id, recipient, content_sent, delivery_status,
provider_message_id, attempts, created_at, last_attempted_at
```

### loan_details

```
id, lead_id, lender_name, application_number, applied_amount,
approved_amount,
```

```
application_status, applied_at, approved_at, disbursed_at, rejection_reason,
notes, updated_at
```

**audit_logs**

```
id, actor_user_id, action, entity_type, entity_id, details_json, ip_address,
created_at
```

---

# 17. API Design Conventions

## 17.1 REST API Standards

All APIs must follow RESTful conventions:

- Base URL: `https://api.{domain}/v1/`
- HTTP methods used as per semantics: GET for retrieval, POST for creation, PUT or PATCH for update, DELETE for deletion.
- HTTP status codes must accurately reflect the result: 200 for success, 201 for creation, 400 for bad request, 401 for unauthorized, 403 for forbidden, 404 for not found, 422 for validation errors, 500 for server errors.
- All API responses must follow a consistent envelope format:

```json
{
  "success": true,
  "data": {},
  "message": "Optional message",
  "error": null,
  "pagination": {
    "page": 1,
    "limit": 20,
    "total": 150
  }
}
```

# 18. Non-Functional Requirements

## 18.1 Performance

- API response time for standard requests must not exceed 500ms at the 95th percentile under normal load.
- Dashboard load time must not exceed 3 seconds.
- The landing page must load within 3 seconds on a 4G mobile connection.

- The system must support a minimum of 500 concurrent users on the web admin portal without degradation.

## 18.2 Availability

- The system must target 99.5% uptime (approximately 43 hours downtime per year).
- Scheduled maintenance windows must be communicated to users in advance and kept under 2 hours.
- A health check endpoint must be provided for all services for load balancer and monitoring integration.

## 18.3 Scalability

- The backend must be stateless to support horizontal scaling.
- The database must be provisioned with read replicas to support read-heavy reporting queries.
- Document storage on S3 scales automatically and requires no additional planning.
- Background jobs (notifications, assignment engine) must use a message queue (Bull with Redis) to handle spikes in lead submission without blocking the API.

## 18.4 Accessibility

- The web admin portal must comply with WCAG 2.1 Level AA standards.
- The mobile application must support system-level font scaling.
- All form fields must have appropriate labels and ARIA attributes.

## 18.5 Internationalization

- The initial version of the application will support English only.
- The architecture must be structured to support additional languages in future, using an i18n framework (react-i18next for web, i18n-js for mobile).
- All date formats must account for IST (Indian Standard Time, UTC+5:30).
- Currency formatting must use INR.

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